
An Eight-Task Pilot Study of Task-Level Predictability for Heterogeneous Decision Primitives

Sudip Niroula*
Independent Researcher
ORCID: 0009-0009-8886-114X

Mandip Pokharel
Independent Researcher
ORCID: 0009-0000-9454-7012

Abstract

When a system must make a bounded decision, the task can be implemented using a rule, a conventional classifier, a typed decision model, or a generative language model. These primitives can differ substantially in quality and operational behavior. A natural question is whether measurable properties of the task, fixed before any primitive is evaluated, can predict those outcomes on an unseen task.

We report an eight-task pilot that freezes the task set, task fingerprint, primitive implementations, and prediction specification before the predictive results are observed. The target is a seven-coordinate profile comprising macro-F1, local training time, local inference time, API cost, p50 and p95 per-example latency, and invalid-output rate. The predictor uses ridge regression with penalty 1 for the learned coordinates; API cost is fixed at zero, and training-time prediction is fit only on classifier rows. The baseline is the training-fold mean profile of each primitive. ESCI is held out as a single e-commerce task; the remaining holdouts remove one task family at a time.

Across the measured task-primitive pairs, quality and validity vary substantially by primitive and task. Under the pinned Qwen2.5-0.5B configuration, the invalid-output rate ranges from 0.023 on VitaminC and 0.025 on SMS Spam to 1.000 on Civil Comments, where every output violates the one-label contract and macro-F1 is therefore 0. The ridge predictor is closer than the primitive-mean baseline on macro-F1 for 5 of 8 held-out tasks and on invalid-output rate for 2 of 8. Under every vector-based policy, both predictors select the TF-IDF logistic-regression classifier. On the ESCI holdout, macro-F1 prediction error decreases from 0.2825 to 0.1024 and invalid-rate error from 0.1199 to 0.0444, without changing the selected primitive.

These eight tasks do not establish whether task fingerprints are generally predictive, whether one primitive is generally preferable, or whether the reported latency is a hardware-normalized measure of speed. They also do not establish that the study has introduced a validated router. Rule, classifier, and Laya measurements were obtained on CPU; Qwen was measured on a GTX 1650 Ti in float16. API cost is zero because no API was called.

Keywords: bounded decisions, task fingerprint, task-level model selection, text classification, typed decision models, generative language models, output validity

1 Introduction

We define bounded decisions here as classification problems with a declared label set: an intent, a binary toxicity label, a binary spam label, a clause type, a yes/no/maybe answer, a fact-checking

*Corresponding author: sudip.785421@memc.tu.edu.np.

label, or a product-relevance label. The same decision task can be implemented using different computational primitives. A rule matches a pattern. A classifier is fit on labeled text. A typed decision model returns a distribution over a label set supplied at inference time. A generative model writes text that a parser then accepts or rejects as a label.

Algorithm selection already treats the choice of procedure as a map from problem features to performance (Rice, 1976; Smith-Miles, 2009). LLM routing already chooses among language models, often with cost in view, and sometimes with a notion of task type (Shnitzer et al., 2023; Ong et al., 2024; Liu et al., 2026a; Li, Zhang, et al., 2026b). Comparisons of encoders or linear classifiers with prompted generative models already report quality together with latency or cost on fixed-label text tasks (Zhang et al., 2025; Gonzalez, 2026; Bucher and Martini, 2024; Vajjala and Shimangaud, 2025). Leave-one-dataset-out selection from training-split meta-features has been studied for abstractive summarization (Bino and Shrivastava, 2026). This pilot does not claim those problems as new.

What we measure is narrower. Each task is scored, under one input string and one label set, by the primitives that the protocol marks as eligible. A fingerprint is computed from the task contract and the training split only. A ridge predictor, specified before these predictive results, maps that fingerprint and a primitive identity to the measured profile. We then ask whether the predicted profile is closer to the measured profile than the training-fold mean of that primitive, and whether any closer prediction changes the primitive selected by a policy that was also specified in advance.

The task is the independent unit for that analysis. The evaluation rows inside a dataset estimate that dataset’s scores. They do not multiply the number of tasks. With eight tasks, three of the five selector-training families contain only one task, so the pilot cannot support a general claim about family-level transfer. We state that limit in the body of the paper.

The resulting empirical picture is as follows. Computational primitives exhibit substantial task-dependent differences in quality and operational behavior, but in this eight-task pilot, a simple pre-deployment task fingerprint did not reliably improve prediction over primitive-specific training-fold mean baselines or alter the resulting computational choice. This does not establish that the hypothesis is false, that the classifier is universally best, or that the study has introduced a router.

2 Research questions

- RQ1. How heterogeneous are quality, validity, latency, and cost across computational primitives on these bounded decision tasks?
- RQ2. How much predictive information about primitive behavior is contained in the fixed pre-deployment fingerprint in this pilot?
- RQ3. Does that fingerprint, through the pre-specified ridge predictor, improve held-out prediction over the primitive-specific mean?
- RQ4. Does a closer predicted profile change the primitive selected by the pre-specified policies?
- RQ5. What failure modes appear when the pinned generative configuration is used as a bounded decision primitive?

RQ2 and RQ3 are about this fingerprint and this predictor. They are not questions about every possible fingerprint. RQ1 and RQ5 are descriptive. RQ4 separates prediction error from the downstream choice.

3 Related work

We organize prior work by the claim it already occupies. We do not treat the search as an exhaustive systematic review. The papers below are ones we retrieved and read far enough to place. Three further retrieved papers, on guarded query routing, a preference platform for LLM routers, and latent-task prompt routing, are listed in the bibliography and are not part of the comparison in this section (Šléher et al., 2025; Lai et al., 2026; Zhang et al., 2026).

Algorithm selection. Rice (1976) formulates selection as a map from problem features to algorithm performance. Smith-Miles (2009) surveys that map as meta-learning across disciplines. Work that uses task features to predict which procedure will perform well falls within this broader framework. Our primitive set is not that survey’s portfolio. The general algorithm-selection formulation is therefore not claimed as a contribution of this pilot.

Routing among language models. Shnitzer et al. (2023) train routers on benchmark datasets and apply them to a new task. The router uses embeddings of the new task’s inputs. GraphRouter (Feng et al., 2024) represents tasks, queries, and LLMs in one graph, predicts performance and cost, and then applies a user preference. TRouter (Liu et al., 2026a) uses synthesized task profiles for cold-start routing among LLMs. LLMRouterBench (Li et al., 2026b) re-evaluates a large set of LLM routers and reports that many of them do not reliably outperform a simple baseline, with a remaining gap to an oracle. RouteLLM, Hybrid LLM, RouterBench, and FrugalGPT select among generative models at query level (Ong et al., 2024; Ding et al., 2024; Hu et al., 2024; Chen et al., 2023). Those systems are neighbors. Their action is to choose which language model to call. This pilot instead includes procedures that need not generate text.

Encoder or classifier versus a generative classifier. TaMAS uses a manual task taxonomy to characterize differences in the relative performance of BERT-like models and LLMs on text-classification tasks; it does not train a held-out selector (Zhang et al., 2025). Other studies compare fine-tuned encoders, linear models, or logistic regression with zero-shot or few-shot generative models and, in some cases, report latency and dollar cost (Bucher and Martini, 2024; Edwards and Camacho-Collados, 2024; Vajjala and Shimangaud, 2025; Gonzalez, 2026). Cascades that send uncertain classifier outputs to an LLM use a signal that exists only after a model has run (de Andrade et al., 2024; Abedini and Heydarnoori, 2025; Kondadadi and Ortega, 2026). Repeating the finding that a trained classifier can outperform a prompted LLM on a fixed label set would not be a new result. We cite these studies rather than treating those established comparisons as new results.

Dataset-level meta-learning. Bino and Shrivastava (2026) extract meta-features from the training split of summarization datasets and use leave-one-dataset-out to predict which pretrained summarizer attains the higher score under a stated metric. Among the retrieved literature, this is the closest methodological precedent: features available without the test set, and the dataset as the holdout unit. Their candidates are abstractive summarizers. They predict a selected system under one metric. They do not predict a vector that includes an invalid-output rate, and they do not include a rule, a conventional text classifier, a typed decision model, and a generative decision model together.

Typed decision models and output constraints. Laya is an open decision-model checkpoint (convaiinnovations, 2026). Public descriptions of Jev and Decision 1.0 treat typed decisions as a software interface (Liu et al., 2026b; Li et al., 2026a). We use Laya as one pinned implementation. We do not treat vendor numbers as results. Separately, constrained decoding can remove structural schema failures of small generative models (Chavan, 2026). Our generative pin does not use that constraint. The invalid-output rates below are rates under exact string match, not rates under grammar-constrained decoding.

Within the literature retrieved for this study, we found no paper that trains on a fixed pre-deployment fingerprint and predicts the seven-coordinate profile of a rule, a TF-IDF logistic regression, Laya, and Qwen2.5-0.5B-Instruct under family holdouts and an e-commerce holdout. This is a statement about the sources we obtained. It is not a claim that the experiment is absent from every index, and it is not a claim of priority.

4 Problem formulation

For a task t and an eligible primitive p , the measured profile is

$$Y_{t,p} = (Q, C_{\text{train}}, C_{\text{infer}}, C_{\text{api}}, L_{p50}, L_{p95}, V). \quad (1)$$

Q is macro-F1 on the frozen evaluation rows. Accuracy is stored and is not the quality coordinate. C_{train} and C_{infer} are local training time and inference time, in seconds. C_{api} is API cost. L_{p50} and L_{p95} are per-example latency in milliseconds, excluding model load. V is the fraction of evaluation rows whose output is not exactly one legal label.

An invalid row is already incorrect for macro-F1. V is kept as its own coordinate because a legal wrong label and an output that is not a label are different failures, and macro-F1 does not say which occurred.

The prediction task is to estimate

$$(\text{fingerprint of } t, \text{ identity of } p) \mapsto \hat{Y}_{t,p}. \quad (2)$$

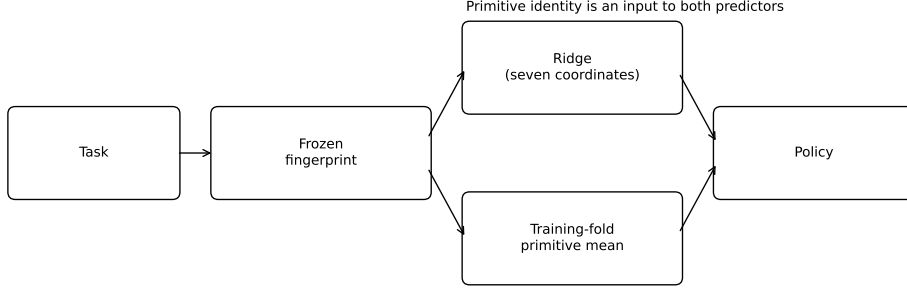


Figure 1: A task yields the frozen fingerprint. Primitive identity enters the ridge-based prediction coordinates and the training-fold primitive mean. A policy reads a predicted profile. There is no arrow from the task directly to a selected primitive.

The fingerprint is shared across primitives. The primitive identity is an intercept shift, with the classifier as the reference level. There is no product between a fingerprint coordinate and a primitive indicator. A policy may read $\hat{Y}$ after it is predicted. The policy is not the training label. This split is deliberate: a single utility would encode a particular preference directly into the prediction target, and encoder-versus-LLM work already varies such weights (Gonzalez, 2026).

Higher Q is better. Lower cost, latency, and V are better. Ineligible pairs have no target. The rule is eligible only on SMS Spam. Measured zeros are retained as zeros. Training time for the rule, Laya, and Qwen is zero because those primitives are not trained on individual tasks. API cost is zero because no API was called.

5 Computational primitives

All eligible primitives on a task see the same input string and the same label set.

Rule. A single regular expression was written and hashed before any scores were measured. It maps an SMS message to spam on a match and to ham otherwise. SHA-256 of the file bytes, including the trailing newline, is

264e288b89a430005de7b72fdfeb77862e46b23b7415c76df285e4e5e1a7f992.

The expression is not fit on labels. No other task receives a rule. A rule is not added to a 77-, 100-, or 151-way label set in response to a lower-scoring primitive.

Classifier. Word unigrams and bigrams, `min_df=2`, at most 30,000 features, sublinear term frequency, and logistic regression with the saga solver, $C = 1$, and at most 200 iterations. The iteration limit was not increased after a convergence warning on SMS Spam. Tasks with at most 100,000 fingerprint training rows use every such row (Cfull). Civil Comments, VitaminC, and ESCI are fit on a stratified 100,000-row subset (C100k). That cap is a frozen compute limit specified before evaluation. It is not a scarcity experiment and it is not the full-data classifier.

Typed decision model. convaiinnovations/laya, English router, one choice question. Criteria keys are the sorted training labels. Descriptions equal those keys. The instruction is “Which label applies?” Context limits are `max_len=2048` and `head_max_len=1024`. Inference was run on CPU. Load time is stored outside latency.

Generative model. Qwen/Qwen2.5-0.5B-Instruct, `do_sample` false, at most 32 new tokens, float16 on a GTX 1650 Ti. The prompt instructs the model to return exactly one label from a sorted list and not to provide an explanation. An output is valid only if, after stripping whitespace and at most one matching pair of surrounding quotes, it equals a legal label. All other outputs are classified as invalid. This configuration was not returned after the invalid-output rates were observed.

6 Task fingerprint

The fingerprint uses the task contract and the training split only. It does not use test labels, test examples, any primitive’s outputs, or any coordinate of Y . It does not use the benchmark name, the dataset identifier, or a subjective difficulty rating.

Token length is `bert-base-uncased` WordPiece without special tokens. The revision is

86b5e0934494bd15c9632b12f734a8a67f723594.

Label-name similarity is mean-pooled `all-MiniLM-L6-v2`. The revision is

1110a243fdf4706b3f48f1d95db1a4f5529b4d41,

after lowercasing and replacing underscores and hyphens with spaces. Normalized entropy is Shannon entropy in nats divided by $\ln K$.

The fingerprint features supplied to the predictor are, in order: $\log_{10}$ of the training-row count, $\log_{10}$ of K , normalized entropy, $\log_{10}$ of the max/min class-count ratio, $\log_{10}$ of the minimum and median rows per class, $\log_{10}$ of the median and P95 input lengths, and the mean and minimum label-name cosines. The relation code is supplied as a binary feature that is 1 for paired-text tasks. Primitive identity is supplied separately through primitive-indicator variables, with the classifier as the reference level. Input structure and evidence required make the same split of these eight tasks as that paired-text indicator, so they are not entered twice. The external-knowledge flag is false for every task. Documented shift is `unspecified` on four selector-training tasks and blank on VitaminC, so it is recorded and not used as a feature. The rule-eligibility flag is true only for SMS Spam and is not used as a fingerprint feature; primitive eligibility is determined by the primitive mask.

For Civil Comments and ESCI, the median and P95 input lengths are computed on a 20,000-row training sample drawn with seed 20260926. Their imbalance uses the full training split. Tables 1a and 1b report the fingerprint values used by the study.

7 Experimental protocol

Evaluation rows are drawn before any fit. If the official test split contains at most 1,000 rows, all rows are used; otherwise, 1,000 rows are sampled without replacement. PubMedQA’s test half has 500 rows, so its evaluation count is 500. Every other task in Table 2 has 1,000 evaluation rows.

The ridge specification was frozen before this fit. Each learned coordinate uses `Ridge(alpha=1)` with an intercept. API cost is fixed at zero, and training-time prediction is fit only on classifier rows because the other primitives are not trained on individual tasks. Numeric fingerprint columns are standardized with the population mean and standard deviation of the training tasks in that fold. A column with zero variance is set to 0, including on the held-out task. Primitive indicators are not standardized. Predictions of Q and V are clipped to $[0, 1]$. Time coordinates are clipped at 0. The penalty parameter was not tuned. No interaction was added after the errors were seen. Training time for the rule, Laya, and Qwen is fixed at zero because these primitives are not trained on individual tasks.

The baseline ignores the fingerprint. For each primitive present in the training tasks, it predicts the arithmetic mean of that primitive’s measured coordinate. An unseen primitive is not predicted and is

not imputed from another primitive. On the security holdout, SMS Spam is removed, so the rule has no training-fold mean and is therefore not predicted.

Task-level error is the mean absolute error across eligible primitives that both predictors can fill. Coordinates are not averaged into one score. A negative difference means the ridge prediction was closer.

Policies run after a vector exists. Always-Qwen, always-classifier, always-Laya, and the mask default do not read a vector. The mask default selects the rule when it is eligible and the classifier otherwise. The remaining policies read either a predicted vector or, for the oracle analysis, the measured vector. The measured vector is not a deployable predictor. The policies are: highest macro-F1; lowest L_{p50} ; equal weights $U = \tilde{Q} - \tilde{C}_{\text{infer}} - \tilde{L}_{p50}$; shared weights $U = \tilde{Q} - \frac{1}{2}\tilde{C}_{\text{infer}} - \frac{1}{2}\tilde{L}_{p50}$. Tildes are min-max maps whose minimum and maximum come from the training tasks of that fit, not from the held-out task. Ties break in the order rule, classifier, decision model, generative model. V is reported with the chosen primitive and is not given a weight.

Family labels are taken directly from the screening record and are not used as features. Intent holds out Banking77 and CLINC150 plus together. Verification holds out PubMedQA and VitaminC together. Moderation, security, and business/legal each hold out one task. For the e-commerce holdout, the predictor is fit on all seven selector-training tasks and evaluated on ESCI. ESCI does not enter any fit, any primitive mean, any standardization, or any utility range. A random 70/15/15 split of tasks is not run. Seven selector-training tasks are insufficient to support a meaningful 70/15/15 task-level split.

Example-level uncertainty concerns variation in a score across the evaluation rows of a single dataset. Task-level uncertainty concerns variation across tasks. Family-level uncertainty would require multiple tasks in each held-out family and several tasks left behind that still represent the fingerprint’s informative features. This design has eight tasks. Three family holdouts are single-task holdouts. The verification holdout removes both paired-text training tasks, so that feature has no variance and is zeroed. We do not report p -values. A count such as “5 of 8” is a description of this sample, not a population rate.

8 Datasets and task families

Tables 1a and 1b are the frozen fingerprint, split into two parts so the type stays readable. All tasks except ESCI are used for selector training. The paired-text relation marks tasks whose input joins two texts. The remaining tasks use a single text field. The releases are those named in the screening record (Casanueva et al., 2020; Larson et al., 2019; Civil Comments; SMS Spam Collection; Chalkidis et al., 2021; Jin et al., 2019; Schuster et al., 2021; Reddy et al., 2022).

Table 1a: Training size, imbalance, and relation. Lengths are in Table 1b. Imbalance uses the full training split. Entropy is normalized entropy. The source file is identified in Section 14.

Task	Family	Train rows	K	Entropy	Max/min	Relation	Selector
Banking77	Intent	10003	77	0.9920	5.343	utterance	yes
CLINC150 plus	Intent	15250	151	0.9990	2.500	utterance	yes
Civil Comments	Moderation	1804874	2	0.4021	11.505	utterance	yes
SMS Spam	Security	4458	2	0.5681	6.467	utterance	yes
LEDGAR	Business / legal	60000	100	0.9348	137.696	utterance	yes
PubMedQA fold 0	Verification	450	3	0.8516	5.082	paired text	yes
VitaminC	Verification	370653	3	0.9030	3.505	paired text	yes
ESCI	E-commerce	1393063	4	0.6228	33.310	paired text	no

Civil Comments is binarized at a toxicity threshold of 0.5. The input is comment text only. PubMedQA training rows are fold 0 of the official seed-0 split, 450 rows, not the undivided 1,000-row pool. ESCI uses the English (US), large version from the official examples file. Product text for the length sample and for model inputs was read from a tasksource copy keyed by product id and locale, because the official products file was not fully downloaded. This provenance is part of the frozen measurement protocol; it does not change the official example-row set. The official examples filter matches 1,393,063 training rows.

Table 1b: Input length and label-name similarity. Lengths for Civil Comments and ESCI are computed on the frozen 20,000-row training sample. Entropy in Table 1a is normalized entropy.

Task	Median tokens	P95 tokens	Mean cosine	Min cosine
Banking77	11.0	35.0	0.2011	−0.1309
CLINC150 plus	9.0	15.0	0.1342	−0.1968
Civil Comments	46.0	201.0	0.5701	0.5701
SMS Spam	18.0	53.0	0.2830	0.2830
LEDGAR	104.0	379.0	0.2194	−0.1086
PubMedQA fold 0	319.5	490.1	0.5272	0.4000
VitaminC	45.0	89.0	0.2113	0.1966
ESCI	213.0	805.0	0.2643	0.1075

9 Results

9.1 Heterogeneity

Table 2 is the performance matrix. Blank primitive–task pairs were not eligible. Missing primitive–task pairs are left blank rather than assigned zero. Cfull and C100k are the classifier. Device is part of the record.

Table 2: Measured profiles. Q is macro-F1. Train and infer are seconds. API cost is zero because no API was called. Latency is milliseconds and does not include model load. Invalid is a count. The rule, classifier, and Laya were run on CPU. Qwen was run on a GTX 1650 Ti in float16. C100k marks the frozen 100,000-row classifier fit. The underlying numerical artifact is provided with the accompanying repository.

Task	Primitive	Q	Time (s)		API (\$)	Latency (ms)		Invalid	n	Device
			Train	Infer		p50	p95			
SMS Spam	Rule	0.8576	0.000	0.028	0	0.020	0.062	0	1000	CPU
SMS Spam	Classifier	0.9393	0.295	0.351	0	0.188	0.719	0	1000	CPU
SMS Spam	Laya	0.7822	0.000	724.349	0	698.724	941.798	0	1000	CPU
SMS Spam	Qwen	0.2067	0.000	308.276	0	323.515	417.066	25	1000	GPU
PubMedQA	Classifier	0.2691	0.635	0.194	0	0.261	0.623	0	500	CPU
PubMedQA	Laya	0.3128	0.000	1019.183	0	1978.849	2842.361	0	500	CPU
PubMedQA	Qwen	0.2327	0.000	474.746	0	932.252	1239.578	174	500	GPU
Banking77	Classifier	0.8508	2.412	0.267	0	0.184	0.456	0	1000	CPU
Banking77	Laya	0.3888	0.000	5260.039	0	5212.193	5888.881	0	1000	CPU
Banking77	Qwen	0.1316	0.000	1360.532	0	1274.864	1668.698	514	1000	GPU
CLINC150 plus	Classifier	0.8299	5.033	0.237	0	0.177	0.337	0	1000	CPU
CLINC150 plus	Laya	0.4280	0.000	4614.129	0	4584.007	4905.017	0	1000	CPU
CLINC150 plus	Qwen	0.2465	0.000	1455.811	0	1336.641	1980.492	745	1000	GPU
LEDGAR	Classifier	0.7418	139.867	0.298	0	0.215	0.481	0	1000	CPU
LEDGAR	Laya	0.2088	0.000	4724.659	0	4528.972	6190.025	0	1000	CPU
LEDGAR	Qwen	0.0281	0.000	1346.729	0	1236.580	1887.759	86	1000	GPU
Civil Comments	Classifier (C100k)	0.6298	19.757	0.316	0	0.196	0.628	0	1000	CPU
Civil Comments	Laya	0.7367	0.000	807.656	0	726.485	1376.470	0	1000	CPU
Civil Comments	Qwen	0.0000	0.000	544.926	0	489.657	808.857	1000	1000	GPU
VitaminC	Classifier (C100k)	0.3826	15.300	0.251	0	0.176	0.416	0	1000	CPU
VitaminC	Laya	0.2165	0.000	797.699	0	766.750	1045.356	0	1000	CPU
VitaminC	Qwen	0.2209	0.000	603.543	0	578.279	779.076	23	1000	GPU
ESCI	Classifier (C100k)	0.2189	77.896	0.253	0	0.181	0.394	0	1000	CPU
ESCI	Laya	0.1527	0.000	1909.519	0	1444.679	4742.117	0	1000	CPU
ESCI	Qwen	0.0358	0.000	1024.658	0	922.247	2060.757	32	1000	GPU

Table 3: Qwen invalid-output rate. Rate is the invalid count divided by the evaluation count n . The rule, the classifier, and Laya have zero invalid outputs on every scored task.

Task	Invalid / n	Rate
VitaminC	23 / 1000	0.023
SMS Spam	25 / 1000	0.025
ESCI	32 / 1000	0.032
LEDGAR	86 / 1000	0.086
PubMedQA	174 / 500	0.348
Banking77	514 / 1000	0.514
CLINC150 plus	745 / 1000	0.745
Civil Comments	1000 / 1000	1.000

Macro-F1 on SMS Spam runs from 0.2067 (Qwen) to 0.9393 (classifier), with the rule at 0.8576 and Laya at 0.7822. On Banking77 the classifier reaches 0.8508 and Qwen 0.1316. On LEDGAR the classifier reaches 0.7418 and Qwen 0.0281. The relative ordering therefore varies across tasks. On PubMedQA, Laya’s macro-F1 is 0.3128 and the classifier’s is 0.2691. On Civil Comments, Laya’s is 0.7367 and the C100k classifier’s is 0.6298. On VitaminC, the classifier (0.3826) is ahead of Qwen (0.2209) and Laya (0.2165), and Qwen’s invalid-output rate is only 0.023. The remaining performance gap is therefore largely attributable to the correctness of legal label predictions rather than output validity. The measurements are not collapsed into a single ranking.

Local inference time and latency are not comparable across devices. Classifier p50 latency is under 0.3 ms on every task. Laya’s p50 on CPU runs from 698.7 ms on SMS Spam to 5212 ms on Banking77. Qwen’s p50 on the GTX 1650 Ti runs from 323.5 ms on SMS Spam to 1336.6 ms on CLINC150 plus. These values characterize the specific deployments used in this study. They do not establish intrinsic speed differences among the primitives. API cost does not vary.

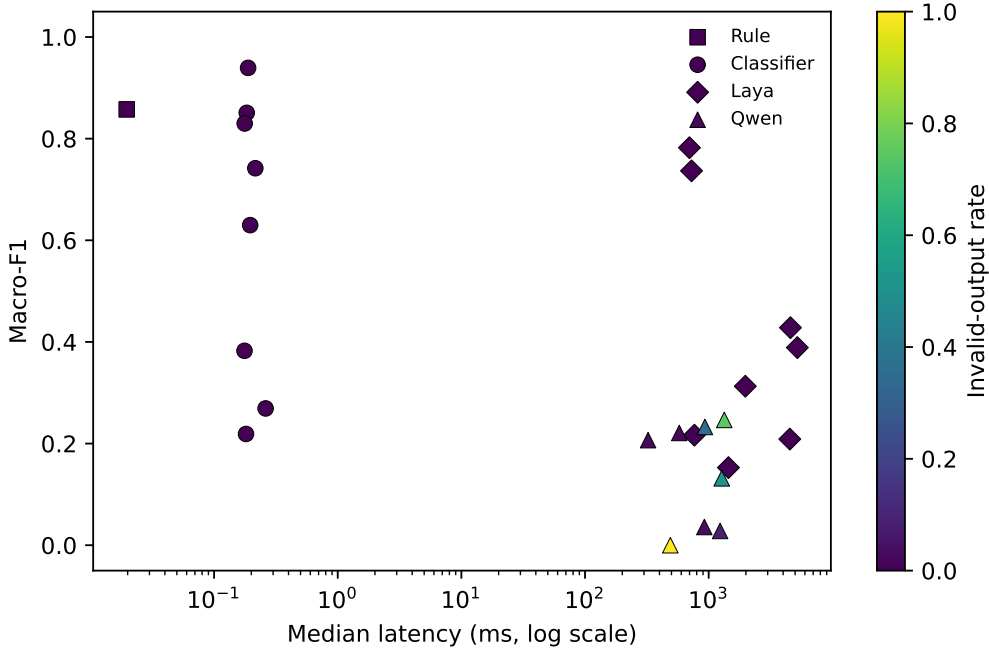


Figure 2: Measured macro-F1 against p50 latency on a log scale, one marker per eligible pair in Table 2. Marker shape is the primitive. Color is the invalid-output rate. The rule, the classifier, and Laya are CPU timings. Qwen is the GTX 1650 Ti in float16. No trend line or confidence band is shown. Because the primitives were measured on different devices, latency should not be interpreted as a hardware-normalized comparison.

Training time is nonzero only for the classifier, from 0.295 s on SMS Spam to 139.867 s on LEDGAR. The C100k fits are 19.757 s (Civil Comments), 15.300 s (VitaminC), and 77.896 s (ESCI).

9.2 Prediction against the primitive mean

Table 4 is the task-level mean absolute error. Each cell averages the eligible primitives both predictors could fill. For the security holdout that set excludes the rule.

Table 4: Held-out absolute error, ridge versus the training-fold primitive mean. Each cell averages the eligible primitives both predictors could fill. For the security holdout that set excludes the rule. The underlying numerical artifact is provided with the accompanying repository.

Holdout	Task	Q ridge	Q mean	V ridge	V mean
Intent	Banking77	0.2112	0.1090	0.1713	0.0725
Intent	CLINC150 plus	0.1623	0.1232	0.2483	0.1495
Verification	PubMedQA	0.2583	0.2785	0.0792	0.0420
Verification	VitaminC	0.2096	0.2688	0.2839	0.1503
Moderation	Civil Comments	0.2075	0.1880	0.3333	0.2366
Security	SMS Spam	0.2271	0.2619	0.2832	0.1426
Business / legal	LEDGAR	0.1562	0.1684	0.1131	0.1188
E-commerce	ESCI	0.1024	0.2825	0.0444	0.1199

Ridge is closer on macro-F1 for PubMedQA, VitaminC, SMS Spam, LEDGAR, and ESCI. It is farther on Banking77, CLINC150 plus, and Civil Comments. Ridge is closer on the invalid-output rate only for LEDGAR and ESCI.

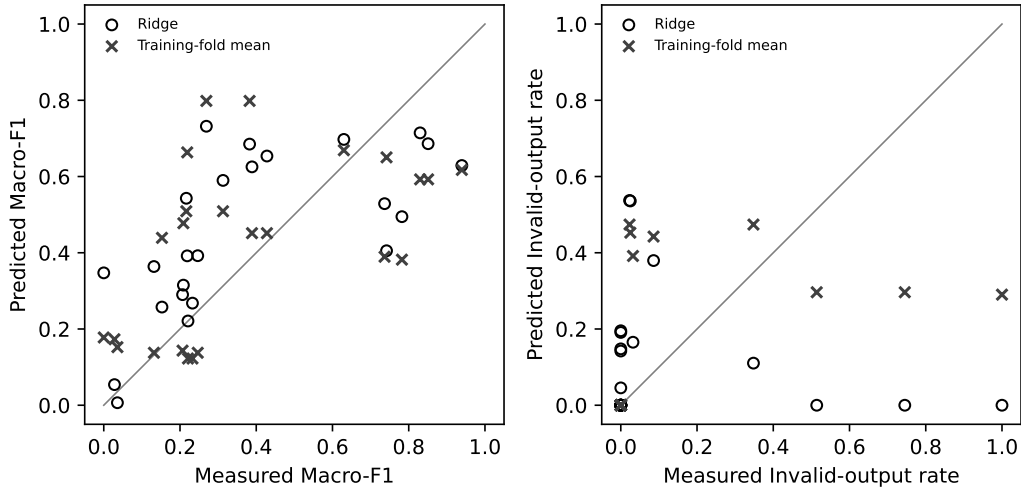


Figure 3: Predicted versus measured macro-F1, and predicted versus measured invalid-output rate, for held-out pairs both predictors filled. The diagonal is equality. Open circles are ridge. Crosses are the training-fold mean.

p50-latency error is smaller for ridge on four tasks and smaller for the mean on four. Those errors are hundreds of milliseconds. They are reported in the result file and are not a hardware-normalized comparison. API-cost error is 0 for both predictors on every task, because every measured value is 0. That tie is not evidence of predictive skill.

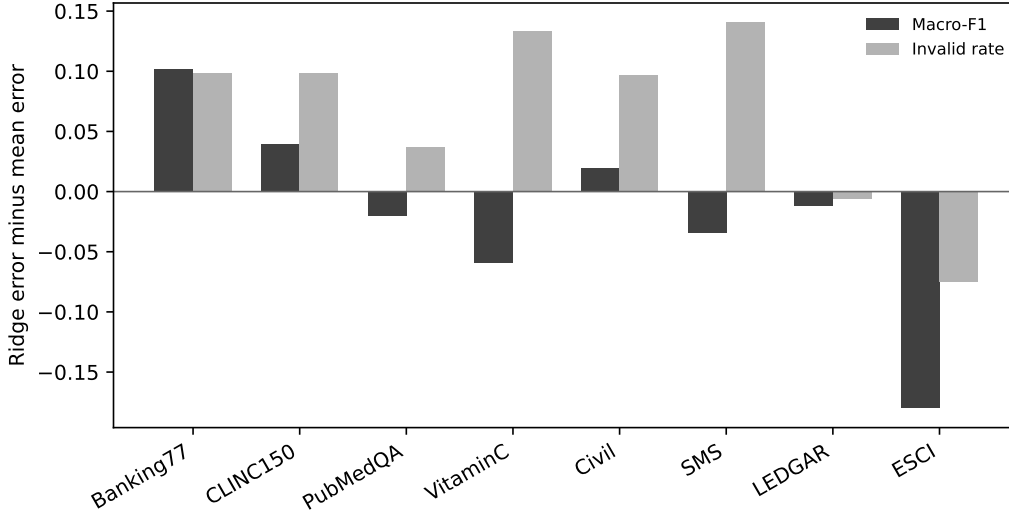


Figure 4: Ridge absolute error minus training-fold-mean absolute error, for macro-F1 and for the invalid-output rate. A negative bar means ridge was closer; the differences are descriptive rather than inferential.

On the fit that uses all seven selector-training tasks, the largest macro-F1 coefficients are the primitive indicators: Qwen -0.385 and Laya -0.134 , against an intercept of 0.593 . The largest fingerprint coefficient on that coordinate is normalized entropy, -0.065 . On the invalid-output rate, the Qwen indicator is $+0.306$ and the largest fingerprint coefficient is mean label cosine, $+0.113$. The fingerprint enters as a shift applied to every primitive. Invalid outputs in this matrix occur for Qwen. A shared shift cannot raise Qwen’s invalid-output rate on a particular kind of task without also raising the classifier and Laya, whose invalid-output rates are 0.

The three largest held-out Qwen invalid-output rates were underestimated by the ridge predictor. Banking77, CLINC150 plus, and Civil Comments have rates 0.514 , 0.745 , and 1 . The unclipped ridge predictions were -0.009 , -0.073 , and -0.039 , and clipping to the $[0, 1]$ interval reduced them to 0. The primitive mean remained near 0.29 . The mean is closer because it retains the average level of Qwen’s observed failure rate. The fingerprint shift removes it.

The verification holdout is also a coverage fact. PubMedQA and VitaminC are the only paired-text tasks in the selector-training pool. Holding them out leaves the relation feature constant, so the column is zeroed. The fit cannot use a structural distinction that the training fold does not contain.

9.3 Choice

On every held-out task, and on all four policies that read a vector, the ridge profile and the primitive mean select the same primitive. In every case, that primitive is the classifier.

The measured oracle, using the same rules on the measured profile, also selects the classifier except in three cases. These cases do not constitute successful predictions, because neither predictor selected the oracle primitive. They show that the objective changes the preferred primitive, and that neither predictor tracked that change.

- Highest macro-F1 on PubMedQA selects Laya: 0.3128 versus 0.2691 for the classifier.
- Highest macro-F1 on Civil Comments selects Laya: 0.7367 versus 0.6298 for the C100k classifier.
- Lowest p50 latency on SMS Spam selects the rule: 0.020 ms versus 0.188 ms for the classifier. The security fold has no rule row in training, so neither predictor can offer a rule.

Equal weights and shared weights select the classifier on every task, including when they read the measured profile. After min-max scaling on the training tasks, Laya’s latency lies at the high end of

the normalized range. The rule–classifier latency difference is small relative to Laya’s normalized latency, and on SMS Spam the classifier has higher macro-F1.

ESCI is the single-task e-commerce holdout. Ridge reduces macro-F1 error from 0.2825 to 0.1024 and invalid-rate error from 0.1199 to 0.0444. Every vector-based policy still selects the classifier, which is also the oracle’s selection. A closer profile did not change the decision. We do not interpret this result as evidence of generalization to the e-commerce domain.

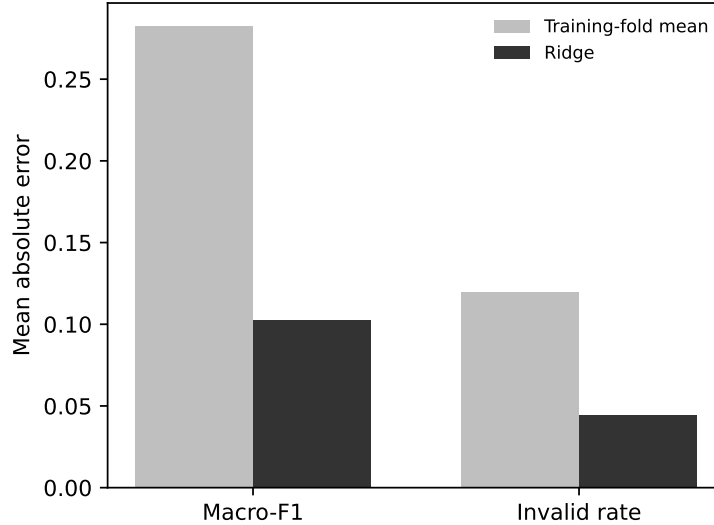


Figure 5: ESCI only. Macro-F1 error is 0.2825 for the training-fold mean and 0.1024 for ridge. Invalid-rate error is 0.1199 and 0.0444. Both predictors and the oracle select the classifier.

9.4 Generative validity

Table 3 is the main validity result. The rule, the classifier, and Laya produce no invalid outputs. Qwen’s invalid-output rate varies substantially across tasks.

On Civil Comments every one of the 1,000 outputs is invalid, so macro-F1 is 0. That zero is a measured contract failure, not a hand-entered score. On Banking77 and CLINC150 plus, invalid outputs are 514 and 745 of 1,000 rows, so the resulting macro-F1 is strongly affected by output validity. On SMS Spam, VitaminC, and ESCI, invalid outputs are 25, 23, and 32 of 1,000 rows. Macro-F1 on those tasks is mostly the legal predictions, and it is still low relative to the classifier (0.2067, 0.2209, and 0.0358).

The raw strings that failed the match were not saved. We cannot say whether they were explanations, extra words, or misspelled labels. We can establish only that they did not satisfy the exact legal-label criterion defined by the frozen protocol. A future experiment could retain the invalid strings to distinguish formatting, explanation, spelling, and other failure modes. Changing the prompt, the decoder, or the parser to reduce V would be a different primitive.

10 Limitations

The task sample contains eight tasks. Intent and verification contain two tasks each. Moderation, security, and business/legal contain one. The e-commerce domain contains one held-out task. Leave-one-family-out in this sample is often leave-one-task-out. Example counts inside Banking77 or Civil Comments do not repair that.

The rule is evaluated on only one task. When that task is held out, the rule cannot be predicted. Only one Laya checkpoint and one Qwen checkpoint are evaluated, so these configurations do not represent their broader model families. Another decision model or another generative model could behave

differently. In particular, grammar-constrained decoding can remove structural invalidity (Chavan, 2026). This study did not use it.

Latency is tied to the device that produced it. A lower Qwen latency is not evidence that generation is cheaper than Laya in general, because Laya ran on CPU. We do not rerun the CPU primitives on GPU after observing the measured matrix. Doing so would replace the frozen measurement with a second condition chosen after the outcome.

API cost is identically zero. The study does not assign a monetary cost to local hardware use and does not compare against a paid API. Training-time comparisons are dominated by the fact that only the classifier is fit on the task.

C100k does not represent full-data training. Civil Comments’ classifier macro-F1 of 0.6298, VitaminC’s 0.3826, and ESCI’s 0.2189 are scores of the capped fits. The fingerprint’s training size still records the full training split.

The fingerprint omits documented shift wherever the code is unspecified or blank. It has no primitive-specific slope. A shuffled-fingerprint control was discussed in an early audit note and was not part of the frozen predictor specification. The included baseline is the primitive-specific training-fold mean. We do not add the shuffled control, or an interaction term, after seeing Table 4.

The literature screen is incomplete. On 26 September 2026, Semantic Scholar’s API returned HTTP 429, and Scopus was not queried. OpenAlex full-text search from 2024 onward was noisy. A closer paper could still sit in an index we did not successfully query.

11 Threats to validity

Construct. Macro-F1, with an invalid row counted as wrong, records whether the bounded decision matched the reference label, including contract failures. V records whether the output was a legal label. Reporting only one of them would mix those questions. We report both. Accuracy is not Q . On Civil Comments the C100k classifier’s accuracy is 0.929 and its macro-F1 is 0.6298, from `research/results/raw/performance_p0_p1.json`. This difference motivates using macro-F1 rather than accuracy as the primary quality coordinate.

Internal. The same evaluation rows and the same input string are used for every primitive on a task. The generative primitive was evaluated on a different device. The seeded subsamples for input length and the C100k classifier fit were specified before evaluation. ESCI product text was obtained from the documented tasksource join rather than from the incompletely downloaded official products file; the row identity remains that of the official examples file. A reader who needs bit-identical product strings must use the documented tasksource join.

External. One generative configuration and one decision-model checkpoint do not represent their families. The label sets are closed. The study does not speak to multilabel decisions, tool use, or open-ended generation. ESCI is one e-commerce task, not a domain sample.

Statistical. The selector’s n is the number of tasks in a fold, not the number of evaluation rows. Holding out intent leaves five training tasks. Holding out a singleton leaves six. Because the number of fingerprint columns exceeds the number of tasks in the fitting folds, the resulting coefficients are descriptive of this fit rather than stable estimates of population-level effects. We do not interpret them as stable partial effects beyond the sign pattern already stated: primitive indicators are larger than individual fingerprint coefficients on the seven-task fit.

12 Discussion

The measured matrix shows substantial task-dependent variation among the evaluated implementations. The classifier’s macro-F1 is high on Banking77, CLINC150 plus, SMS Spam, and LEDGAR, and much lower on PubMedQA, VitaminC, and ESCI. Laya exceeds the classifier on PubMedQA and on Civil Comments under macro-F1, and is far from it on Banking77 and LEDGAR. On SMS Spam, the rule reaches macro-F1 0.8576 compared with 0.9393 for the classifier; this is the only task on which a rule is evaluated. Qwen has the lowest measured macro-F1 among eligible primitives on

seven of the eight tasks. On VitaminC it is 0.2209 and Laya is 0.2165. The source of the performance gap differs across tasks. When V is near zero, the remaining gap primarily reflects errors among legal labels; when V is near one, output-contract failure dominates.

The frozen predictor specification does not allow Qwen to have a primitive-specific fingerprint slope. Under that restriction, the primitive-mean baseline is strong because it already carries the level of each primitive. The fingerprint sometimes moves macro-F1 error down. It usually moves invalid-output-rate error up, including on the tasks where the contract failure is largest. Even the helpful ESCI movement does not change the policy. LLMRouterBench reports a related pattern inside LLM routing: a method can fail to outperform a simple baseline even when model behavior is heterogeneous (Li, Zhang, et al., 2026b). We cite that as a caution, not as the same experiment.

The study does not establish that a fingerprint with primitive-specific slopes, a larger task sample, or a different generative configuration would fail. It also does not establish that such a specification would succeed. The protocol was frozen before Table 4 existed. Changing the model after observing the prediction errors would compromise that pre-specification.

13 Conclusion

Computational primitives exhibit substantial task-dependent differences in quality and operational behavior. In this eight-task pilot, a simple pre-deployment task fingerprint did not reliably improve prediction over primitive-specific training-fold mean baselines, and it did not alter the resulting computational choice. The invalid-output rate of the pinned generative configuration is itself a large, task-dependent part of that behavior, from a few percent of rows to every row. Eight tasks, several of them alone in their family, do not establish whether task fingerprints are generally predictive. They establish only what this frozen measurement found.

14 Reproducibility

Freeze identifiers are FPS-1, FPS-2, EP-1, ST-1, and SD-1, dated 26 September 2026. The fingerprint manifest, performance matrix, and selector results are

```
research/docs/numerical_fingerprint_manifest.md
research/docs/performance_matrix.md
research/docs/selector_results.md
```

The matrix is generated from

```
research/results/raw/performance_p0_p1.json
research/results/raw/performance_dg.json
```

The selector fit and its output are

```
research/src/selector/fit_sd1.py
research/results/raw/selector_sd1.json
```

The figures are generated by `research/paper/make_figures.py`, which reads those JSON files and does not refit the model. The SMS rule file and its hash are in Section 5. Tokenizer and encoder revisions are in Section 6. Raw benchmark corpora are not redistributed with this repository. ESCI product-text provenance is stated in Section 8.

The public repository is <https://github.com/imsudip45/task-level-predictability>.

15 Author contributions

Sudip Niroula: Conceptualization, Methodology, Software, Investigation, Formal Analysis, Data Curation, Visualization, Writing – Original Draft, and Writing – Review & Editing. Mandip Pokharel: Conceptualization, Methodology, Investigation, and Writing – Review & Editing. Both authors reviewed and approved the final manuscript.

16 Conflict of interest

The authors declare no conflict of interest.

References

- Yasaman Abedini and Abbas Heydarnoori. Hybrid LLM routing for efficient app feedback classification, 2025. arXiv:2507.08250.
- Dhivya Bino and Manish Shrivastava. Dataset-aware automated model selection for abstractive summarization: A meta-learning approach. In *Proceedings of COMPUTATIA 2026*, 2026. doi: 10.2991/978-94-6239-713-2_43.
- Martin Juan José Bucher and Marco Martini. Fine-tuned ‘small’ LLMs (still) significantly outperform zero-shot generative AI models in text classification, 2024. arXiv:2406.08660.
- Iñigo Casanueva, Tadas Temčinas, Daniela Gerz, Matthew Henderson, and Ivan Vulić. Efficient intent detection with dual sentence encoders, 2020. arXiv:2003.04807. Banking77 release used here.
- Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. LexGLUE: A benchmark dataset for legal language understanding in English, 2021. arXiv:2110.00976. LEDGAR configuration used here.
- Akash Chavan. Constrained decoding eliminates structural failures in small LLMs but reveals a scale-dependent semantic gap, 2026. arXiv:2609.23742.
- Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance, 2023. arXiv:2305.05176.
- Civil Comments. Civil comments dataset. CC0 release used here: train 1,804,874; validation 97,320; test 97,320. No paper or publication year is identified for this dataset.
- convaiinnovations. Laya model card, 2026. URL <https://huggingface.co/convaiinnovations/laya>.
- Claudio M. V. de Andrade, Washington Cunha, Davi Reis, Adriana Silvina Pagano, Leonardo Rocha, and Marcos André Gonçalves. A strategy to combine 1stGen transformers and open LLMs for automatic text classification, 2024. arXiv:2408.09629.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Ruhle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing, 2024. arXiv:2404.14618.
- Aleksandra Edwards and Jose Camacho-Collados. Language models for text classification: Is in-context learning enough?, 2024. arXiv:2403.17661.
- Tao Feng, Yanzhen Shen, and Jiaxuan You. GraphRouter: A graph-based router for LLM selections, 2024. arXiv:2410.03834.
- Alberto Andres Valdes Gonzalez. Cost-aware model selection for text classification: Multi-objective trade-offs between fine-tuned encoders and LLM prompting in production, 2026. arXiv:2602.06370.
- Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. RouterBench: A benchmark for multi-LLM routing system, 2024. arXiv:2403.12031.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William W. Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering, 2019. arXiv:1909.06146. Fold 0 of the official split.
- Rishik Kondadadi and John E. Ortega. L2D-clinical: Learning to defer for adaptive model selection in clinical text classification, 2026. arXiv:2604.13285.

- Guannan Lai, Haoran Hu, and Han-Jia Ye. RouteJudge: An open platform for reproducible and preference-aware LLM routing, 2026. arXiv:2606.18774.
- Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. An evaluation dataset for intent classification and out-of-scope prediction, 2019. arXiv:1909.02027. Plus configuration, the CLINC150 release used here.
- Delong Li, Xu Wang, Haochen Gong, Rui Lang, and Guangsheng Yu. Replacing large language models with Jev decision models for low-latency edge service orchestration, 2026a. arXiv:2609.22753.
- Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang, Shengji Tang, Qiaosheng Zhang, Yang Chen, Biqing Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu. LLMRouterBench: A massive benchmark and unified framework for LLM routing. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 37733–37754, 2026b. doi: 10.18653/v1/2026.findings-acl.1881. URL <https://aclanthology.org/2026.findings-acl.1881/>.
- Hui Liu, Bin Zou, Kecheng Chen, Jie Liu, Wenya Wang, and Haoliang Li. Task-aware LLM routing with multi-level task-profile-guided data synthesis for cold-start scenarios. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22047–22076, 2026a. doi: 10.18653/v1/2026.acl-long.1007. URL <https://aclanthology.org/2026.acl-long.1007/>.
- Xunzhuo Liu, Haichen Zhang, Mahdi Ghodsi, Andy Luo, Adam Kovacs, Huamin Chen, Xue Liu, and Bowei He. Decision 1.0: Towards open decision foundation models, 2026b. URL <https://vllm-sr.ai/decision-paper.pdf>. Manuscript retrieved 26 September 2026. No arXiv identifier was on the retrieved copy.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data, 2024. arXiv:2406.18665.
- Chandan K. Reddy, Lluís Màrquez, Fran Valero, Nikhil Rao, Hugo Zaragoza, Sambaran Bandyopadhyay, Arnab Biswas, Anlu Xing, and Karthik Subbian. Shopping queries dataset: A large-scale ESCI benchmark for improving product search, 2022. arXiv:2206.06588. English (US), large version, as filtered here.
- John R. Rice. The algorithm selection problem. *Advances in Computers*, 1976. doi: 10.1016/S0065-2458(08)60520-3.
- Tal Schuster, Adam Fisch, and Regina Barzilay. Get your vitamin C! robust fact verification with contrastive evidence, 2021. arXiv:2103.08541. Fact-verification release used here.
- Tal Shnitzer, Anthony Ou, Mfrian Silva, Kate Soule, Yuekai Sun, Justin Solomon, Neil Thompson, and Mikhail Yurochkin. Large language model routing with benchmark datasets, 2023. arXiv:2309.15789.
- Richard Šléher, William Brach, Tibor Sloboda, Kristián Košťál, and Lukas Galke. Guarded query routing for large language models. In *ECAI 2025, Frontiers in Artificial Intelligence and Applications*. IOS Press, 2025. doi: 10.3233/FAIA251304. arXiv:2505.14524.
- Kate Smith-Miles. Cross-disciplinary perspectives on meta-learning for algorithm selection. *ACM Computing Surveys*, 2009. doi: 10.1145/1456650.1456656.
- SMS Spam Collection. SMS spam collection dataset. URL <https://doi.org/10.24432/C5CC84>. UCI release used here. No publication year for a paper is identified for this dataset.
- Sowmya Vajjala and Shweta Shimangaud. Text classification in the LLM era – where do we stand?, 2025. arXiv:2502.11830.
- Junyan Zhang, Yiming Huang, Shuliang Liu, Yubo Gao, and Xuming Hu. Do BERT-like bidirectional models still perform better on text classification in the era of LLMs? In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 18980–18989, 2025. doi: 10.18653/v1/2025.findings-emnlp.1033. URL <https://aclanthology.org/2025.findings-emnlp.1033/>.

Yunyi Zhang, Soji Adeshina, Sheng Guan, Ashwin Ganesh, Zhen Han, Vassilis N. Ioannidis, Huzefa Rangwala, and George Karypis. Scalable prompt routing via fine-grained latent task discovery, 2026. arXiv:2603.19415.